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The Completion

**A relativistic field theory carrying $a_0 = \kappa c \sqrt{(G\rho - \Lambda)}$ **

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1. The action

$$S = \int d^4x \sqrt{-g} \left[\underbrace{\frac{R - 2\Lambda_{\text{bare}}}{16\pi G}}_{\text{Einstein-Hilbert}} + \underbrace{\mathcal{L}_{\text{aether}}[A^\mu, g]}_{\text{unit-timelike } A^\mu A_\mu = -1} + \underbrace{\mathcal{F}(Y, Q)}_{\text{the dark sector}} \right] + S_{\text{matter}}[g, \psi]$$

with the two scalar invariants built from the khronon φ and the aether A^μ ,

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi.$$

This is **Aether-Scalar-Tensor** (Skordis & Złošnik 2021, PRL **127** 161302) — chosen because it is the only relativistic MOND-class theory that reproduces the CMB. The framework’s content is entirely in the choice of the free function , which **factorises**:

$$\mathcal{F}(Y, Q) = \underbrace{\frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right)}_{\text{MOND}} + \underbrace{K(Q)}_{\text{dark energy + dark matter}}$$

1.1 The Y-sector — MOND

Y is fixed by the framework’s Route A kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$, $y = g_{\text{bar}}/a_0$. In AQUAL variables it is the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = g_{\text{obs}}/a_0.$$

with the normalisation

$$a_0 = \kappa c \sqrt{G \rho_\Lambda} = \frac{c H_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ m s}^{-2}, \quad \kappa = \frac{1}{2}, \quad Z = 2\sqrt{8\pi/3}.$$

$\kappa = \frac{1}{2}$ is **FITTED**, not derived. See §4.

1.2 The Q-sector — an offset DBI khronon

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D$$

Three jobs from one function:

piece	limit	what it is
the offset $-M^4$	$u = 0, K = 0$	$\rho = -K = M^4, p = K = -M^4$ $\Rightarrow \mathbf{w} = \mathbf{-1}$ exactly : dark energy
the deviations	$u \ll \Lambda_D$	$K \rightarrow \mu^2 u^2 / 2 \Rightarrow w \rightarrow 0$: dust , i.e. the dark matter
the saturation	$u \rightarrow \Lambda_D$	K bounded, $K \rightarrow \infty \Rightarrow w \rightarrow 0$ again: kills the early stiff phase

Parameters: $\mathbf{M}$ (vacuum scale, $M^4 = \rho_- \Lambda$), μ (Helmholtz mass), $**\Lambda_D**$ (DBI field-space scale), $\mathbf{I}_0$ (the conserved shift charge = the dust amount).

2. What is verified

#	property	value / statement	script
1	Lensing — the test that killed modified inertia	$\Phi = \Psi \Rightarrow \gamma_{\text{PPN}} = 1, M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly. $21.2\sigma \rightarrow 0.601\sigma$	<code>mi_relativistic_completion_aest</code>
2	Kernel embeds in $_{-Y}$	deep-MOND $\mu \rightarrow x$ exact; Newtonian $\$ \rightarrow 1; x(u) \text{bijection}(h = e^{(-u)}(1+u) > 0); \text{free function} *$ $*\text{convex} ** \text{same} 3 *$ $*\text{Newtonian residual} *$ $* e^{(-\sqrt{\$}y)} = 3.6 \times 10^{-3457}$ at Earth's orbit	same

#	property	value / statement	script
4	Dark energy is not an input	$w = -1$ exactly at the condensate minimum — proved, not assumed	mi_condensate_vacuum_energy_a0_2
5	Ghost-free over the whole field range	$K = \mu^2(1-s^2)^{-3/2} > 0$ for all $ s < 1$	mi_dbi_khronon_2026.py
6	Subluminal everywhere	$c_s^2 = \Lambda_D$ $s(1-s^2)/(1+\Lambda_D s) \leq 0.385 \Lambda_D$	same
7	The published no-go is dissolved	Blanchet & Skordis 2024 §4.3.1's 455× conflict reverses sign : bounded pressure $\Rightarrow w \rightarrow 0$ early $\Rightarrow$ cosmology becomes a <i>lower</i> bound on μ^{-1} , same direction as MOND	same
8	Theorem : why no polynomial works	for $K \sim u^n$, early-time $w \rightarrow 1/(n-1)$. Quadratic 1, quartic $\frac{1}{3}$, sextic — all fail. Only boundedness gives $w \rightarrow 0$	same
9	CMB acoustic peaks — real CLASS run	at $\Lambda_D \leq 10^{-2}$, $c_s^2(\text{recomb}) = 2.9 \times 10^{-8}$; TT unchanged to 0.069% , $P(k=0.2)$ to 1.71%. Indistinguishable from CDM	mi_dbi_cmb_class_run_2026.py
10	BTFR	exact, from convexity	mi_route_a_field_theory_2026.py
11	g^{-2} Lorentz violation	restored by the aether (pure Bekenstein–Milgrom had lost it)	mi_relativistic_completion_aest
12	a_0 tied to Λ	$a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ with $m_{\text{cond}} = M^2/(\sqrt{2}M_{Pl}), M^{\{4\}} = \rho \Lambda$ — agrees to 0.076%	mi_condensate_vacuum_energy_a0_2

#	property	value / statement	script
13	Clusters	primordial shift-charge overdensity; dissolves the 1403× fork and Mistele–McGaugh– Hossenfelder’s 2500× tension ($\mu^2 =$ 5.2×10^{-8} satisfies their weak-lensing bound by 1.9×10^4)	mi_shift_charge_ic_route_2026.py
14	Directional EFE signal	pure MI predicted <i>exactly zero</i> ; AQUAL-class predicts 1–4% signed; first firing gave $\hat{A} = +2.95$, $p = 0.029$, with the AQUAL sign	mi_dr4_anisotropy_and_gated_2026.py

3. Parameter count, honestly

	Λ CDM	this completion
dark sector	Ω_c, Ω_Λ	$M (= \rho_\Lambda^{1/4}), I_0, \mu, \Lambda_D$
MOND scale	—	a_0 — not independent , = $m_{\text{cond}}/(4\sqrt{\pi})$ via item 12
bounds	—	Λ_D : $1.9 \times 10^{-10} \ll \Lambda_D \lesssim$ 10^{-2} (~ 7.7 orders, CMB-bounded); $\mu^{-1} = 4392$ Mpc (forced by item 12); $I_0 \approx$ Ω_{dm} (an IC)

So: **four dark-sector numbers against Λ CDM’s two**, with a_0 derived from one of them rather than added. Not fewer parameters. What it buys is that Λ , dark matter and MOND come from **one function** instead of three unrelated sectors.

4. What is NOT claimed — read this before quoting anything above

1. $\kappa = \frac{1}{2}$ **IS NOT DERIVED**. Item 12 looks like a derivation and is not. $a = m_{\text{cond}}/(4\sqrt{\pi})$ is *algebraically identical* to $a = \frac{1}{2}\mathcal{V}(G_\Lambda)$ and to $Z = 2\mathcal{V}(8/3)$ — three costumes, one statement — and the residue is **convention-dependent** ($4\sqrt{\pi}$ with the reduced Planck mass, $\frac{1}{2}$ with the non-reduced). A derived number cannot depend on a bookkeeping choice. It

is a **relabelling**, as this corpus’s own theorem already established. What genuinely moved: “why $\kappa = \frac{1}{2}$?” becomes “why is $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$?”, which is sharper because m_{cond} is a *derived* quantity of a real field theory rather than a definition.

2. **Dark matter EXISTS**, at the full Ω_{dm} . Removing the pressureless component moves H_3/H_1 by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). What does **not** exist is a dark-matter *particle*: it is the Q-sector of the same scalar that supplies Λ and MOND, and by item 13 it is absent where rotation curves are measured. **The defensible slogan is “no dark matter particle, and none in galaxies” — never “no dark matter.”**
3. **The post-recombination growth history is NOT verified.** Item 9 covers the acoustic peaks only. Holding c_s^2 at its peak value for all time *destroys* $P(k=0.2)$, so the pass depends essentially on the bump being transient (it peaks at $z \approx 189$). **A patched CLASS fluid carrying $c_s^2(a)$ is required, not optional.**
4. **Item 13’s initial condition is not confronted with Lyman- α .** It needs a khronon transfer function falling to $T \approx 0.33$ at $k \approx 4.5 \text{ Mpc}^{-1}$, which in a ΛCDM -style analysis would be excluded. Whether the MOND Y-sector’s compensation rescues it is **uncomputed, and is the largest single owed item in the programme.**
5. **Cassini Q_2 is inherited** ($3\text{--}15\sigma$) and is not relieved by anything here.
6. **PPN preferred-frame parameters α_1, α_2** for this have not been computed against lunar-laser and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote only.** $\gamma_{\text{v}} = \sqrt{\nu(y_{\text{extN}})}$; the full nonlinear AQUAL-EFE solve is owed.
8. **This is not a theory of everything** and no part of it addresses the Standard Model. The 2026-06-23 retraction stands.

5. The falsifiable predictions

prediction	value	how to test
wide-binary boost	$\gamma_{\text{v}} = \sqrt{\nu(y_{\text{extN}})}$	Gaia DR4 , pre-registered and hash-stamped
directional EFE asymmetry	1–4%, signed	already firing at $p = 0.029$ with the right sign
cluster-to-cluster scatter in the residual	$\sigma(\xi)/\xi \approx 0.02\text{--}0.60$	existing cluster samples — separates item 13 from the R^2 lever, which predicts zero
cluster mass profile misfit	ρ_{c} flat, $M_{\text{c}} \sim r^3$, can go negative	fit clusters with the khronon profile instead of NFW
g^{-2} Lorentz violation	computable $s_{\mu\nu}$	SME sector bounds
accelerated structure formation	earlier massive objects	JWST high- z massive galaxies (McGaugh et al. 2024)

6. Credit

$\nu = \sqrt{1+1/y}$ is **Milgrom 1999 PLA 253:273 eq 9**. MOND: Milgrom 1983 ApJ 270:365. AQUAL: Bekenstein & Milgrom 1984 ApJ 286:7. TeVeS: Bekenstein 2004 PRD 70:083509. **AeST: Skordis & Złóśnik 2021 PRL 127:161302** — the completion itself. Ghost condensate: Arkani-Hamed, Cheng, Luty & Mukohyama 2004 JHEP 0405:074; nonlinear: +Wiseman 2007 JHEP 0701:036. $K(Q) = \mu^2(Q-1)^2$ and the ghost- condensate identification: Verwayen, Skordis & Złóśnik 2024; **Blanchet & Skordis 2024 JCAP 11(2024)040** — including §4.3.1, the no-go this work dissolves. Quasi-static AeST: Durakovic & Skordis 2024 JCAP 04:040; Verwayen, Skordis & Boehm 2024 MNRAS 531:272. **Mistele, McGaugh & Hossenfelder 2023 A&A 676:A100** — the weak-lensing challenge item 13 resolves. GDM bound: Kopp, Skordis, Thomas & Ilić 2018 PRL 120:221102. Ly- α floor: Rogers & Peiris 2021 PRL 126:071302. RAR scatter: Desmond 2023. Accelerated structure formation: McGaugh, Schombert, Lelli & Franck 2024.

The distinctive content of this work is the coefficient — $a_0 = \kappa c \sqrt{(G\rho_{\text{--}}\Lambda)}$ with $\kappa = \frac{1}{2}$ — its embedding in $\text{--}Y$, the offset-DBI Q-sector, and items 7, 8, 9, 12 and 13. Everything else is cited above and is not mine.